ALKENE

101. Which is the most reactive hydrocarbon in the following

(a) Ethane (b) Ethyne

(c) Ethene (d) Methane

102. Shape of ethylene molecule is

(a) Tetrahedral (b) Pyramidal

(c) Planar (d) Linear

103. Electrophilic addition on a carbon-carbon double bond involves the intermediate formation of a more stable carbocation. This statement is called

(a) Saytzeff's rule

(b) Baeyer's effect

(c) Markownikoff's rule

(d) None of these

104. $CH_2 = CHCl$ reacts with *HCl* to form

(a) $CH_2Cl - CH_2Cl$

(b) $CH_3 - CHCl_2$

(c) $CH_2 = CHCl.HCl$

(d) None of these

105. Deviation from Markownikoff's rule occurs in presence of

(a) Zinc (b) Peroxides

(c) $Hg - Zn/HCl$ (d) All of these

106. Presence of peroxides affects the addition of

(a) *HBr* (b) *HCl*

(c) *HI* (d) All of these

107. Catalyst used in dimerisation of acetylene to prepare chloroprene is

(a) $HgSO_4 + H_2SO_4$

(b) Cu_2Cl_2

(c) $Cu_2Cl_2 + NH_4Cl$

(d) $Cu_2Cl_2 + NH_4OH$

108. Chloroprene is

(a) 2-chloro-1, 3-butadiene

(b) 3-chloro-2, 3-butadiene

(c) 2, 3-dichlorobutadiene

(d) None of these

109. Chloroprene is used in making

(a) Synthetic rubber (b) Plastic

(c) Petrol (d) All of these

110. When isobutyl magnesium bromide in dry ether is treated with absolute ethyl alcohol, the products formed are

(a) $CH_3 - \underset{\underset{CH_3}{|}}{CH} - CH_2OH$ and

CH_3CH_2MgBr

(b) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_2-CH_3$ and

$Mg(OH)Br$

(c) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_3$ and

CH_3-CH_2OMgBr

(d) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_3, CH_2=CH_2$ and

$Mg(OH)Br$

111. The predominant product formed, when 3-methyl-2-pentene reacts with *HOCl*, is

(a) $CH_3-CH_2-\overset{\overset{Cl}{|}}{\underset{\underset{CH_3}{|}}{C}}-\overset{\overset{OH}{|}}{CH}-CH_3$

(b) $CH_3-CH_2-\overset{\overset{Cl}{|}}{\underset{\underset{CH_3}{|}}{C}}-\overset{\overset{Cl}{|}}{CH}-CH_3$

(c) $CH_3-CH_2-\overset{\overset{OH}{|}}{\underset{\underset{CH_3}{|}}{C}}-\underset{\underset{Cl}{|}}{CH}-CH_3$

(d) $CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-\overset{\overset{OH}{|}}{CH}-CH_3$

112. Which of the following occurs easily in ethylene

(a) Addition

(b) Substitution

(c) Elimination

(d) Rearrangement

113. How many *gm* of bromine will react with 21 *gm* C_3H_6

(a) 80 (b) 160

(c) 240 (d) 320

114. Conjugate double bond is present in

(a) Propylene (b) Butadiene

(c) Isobutylene (d) Butylene

115. On passing vapours of an organic liquid over finely divided *Cu* at $573\,K$ the product was an alkene. This reaction is

(a) Catalytic oxidation of primary alcohol

(b) Catalytic dehydrogenation of secondary alcohol

(c) Catalytic dehydrogenation of tertiary alcohol

(d) Catalytic dehydration of tertiary alcohol

116. The total number of sigma $_\sigma$ and pi (π) bonds in an ethylene molecule are

(a) $4\sigma, 2\pi$ (b) $4\sigma, 1\pi$

(c) $5\sigma, 2\pi$ (d) $5\sigma, 1\pi$

117. Cyclic hydrocarbon molecule A has all the carbon and hydrogens in a single plane. All the carbon-carbon bonds are of same length and less that 1.54 $\mathring{A}$ but more than 1.34 $\mathring{A}$. $C-C-C$ bond angle will be

(a) 120° (b) 180°

(c) 100° (d) $109^o 28'$

118. General formula of alkenes is

(a) C_nH_{2n} (b) C_nH_{2n-2}

(c) C_nH_{2n+2} (d) C_nH_{2n-1}

119. The product of following reaction is

$$CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH=CH_2 \xrightarrow[\text{(ii) } NaBH_4]{\text{(i) } Hg(OAc)_2;H_2O}$$

(a) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_2}{|}}{C}}-\underset{\underset{OH}{|}}{CH}-CH_2$

(b) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_2-CH_2OH$

(c) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{OH}{|}}{C}}-\overset{\overset{CH_3}{|}}{CH}-CH_3$

(d) $HOCH_2-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_2-CH_2$

120. Which one of the following organic compounds decolourizes an alkaline $KMnO_4$ solution

(a) CS_2 (b) C_3H_6

(c) C_3H_8 (d) CH_3OH

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